How Generative AI Works

Mechanics Made Intuitive — Day 2

Ever wondered what's actually happening inside ChatGPT when it writes a paragraph, answers a question, or makes up a "fact"? Today we pull back the curtain. No PhD required — just curiosity and a willingness to think in analogies.



Today's Roadmap

Four Big Ideas We'll Cover

01

Tokens, Context & Prediction

How AI reads and generates language, one chunk at a time

02

Transformers — Why They Matter

The architecture that changed everything about modern AI

03

Training Process Overview

From raw internet text to a helpful, instruction-following model

04

Model Behavior & Limitations

Hallucinations, bias, and why AI can be confidently wrong

PART 1

Tokens, Context & Prediction

Before an AI model can understand or generate a single word, it needs to break language down into digestible pieces. These pieces are called **tokens** — and they're the fundamental unit of everything that follows.



What Is a Token?

Think of tokens as **Lego bricks for language**. A model doesn't read letter-by-letter or word-by-word — it reads in chunks. The word **"unbelievable"** might become **["un", "believ", "able"]** — three tokens. Common short words like "the" or "is" are usually one token each.

Why does this matter?

Every model has a **token limit** per session. GPT-4 handles ~128,000 tokens — roughly a 300-page novel. Exceed that, and the model literally forgets what was said earlier, like a conversation that falls off the edge of a cliff.

The Context Window

The context window is the model's **working memory** — everything it can "see" at once. A longer context window means the model can consider more of your conversation, document, or code before generating a response. Older models had tiny windows (~4K tokens). Modern ones are dramatically larger.

Next-Token Prediction: The Core Trick

Here's the surprising truth: **all of generative AI boils down to one task** — predicting the next token, over and over again.

1

Input Prompt

"The capital of France is..."

2

Model Predicts

What token is most likely to come next? → "Paris"

3

Repeat

Now predict the token after "Paris" — and so on until done

> It's like a very sophisticated autocomplete — the kind on your phone keyboard, but trained on almost all of human writing and scaled up by billions of parameters.

PART 2

Transformers — Why They Matter

The transformer architecture, introduced in a landmark 2017 paper titled *"Attention Is All You Need"*, completely redefined what AI could do with language. But to appreciate why, we first need to see what came before.



From RNNs to Transformers

🐢 The Old Way: RNNs

Recurrent Neural Networks read text like a person reading a very long sentence — **one word at a time, left to right**. By the time they reached the end of a long paragraph, they'd often "forgotten" what was at the beginning. They also couldn't be parallelized, making training painfully slow.

⚡ The New Way: Transformers

Transformers read the **entire input at once** and use a mechanism called **self-attention** to decide which words are most relevant to each other — regardless of distance. "The trophy didn't fit in the suitcase because *it* was too big." Attention tells the model that "it" refers to the trophy, not the suitcase.

ⓘ Self-attention is like being able to re-read the whole document instantly and highlight exactly which words matter most for understanding the current word — every single time.

Why Transformers Power Modern AI



Parallelizable Training

Unlike RNNs, transformers process all tokens simultaneously. This means training can be massively distributed across thousands of GPUs — cutting weeks of compute time down to days.



Long-Range Understanding

Attention connects distant words effortlessly. A reference at the end of a 10,000-word document can still relate to context from the very beginning.



Scalability

More data + more parameters = better performance. Transformers scale in a way older architectures never could, which is exactly why GPT, Gemini, and Claude all use them.



PART 3

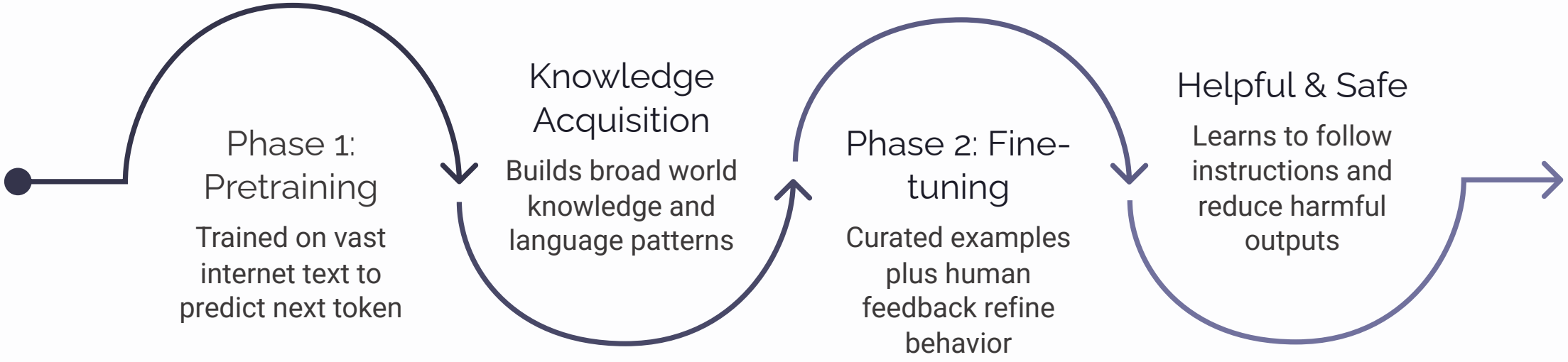
Training Process Overview

A model doesn't arrive pre-loaded with knowledge. It learns from data — enormous amounts of it. Understanding the two-phase training process helps explain both what models know and how they learn to be helpful.

Phase 1: Pretraining — Reading the Internet

During pretraining, the model is exposed to **hundreds of billions of tokens** of text — books, websites, code, scientific papers, Wikipedia, forums, and more. Its only job: predict the next token. Over trillions of such predictions, it learns grammar, facts, reasoning patterns, and even social nuance — all as a side effect of getting better at that one task.

> Think of it like a student who reads every book in the world's largest library before ever attending a class. They absorb patterns, facts, and ways of thinking — without being explicitly taught any single thing.



Phase 2: Fine-Tuning — Learning to Be Helpful

Instruction Tuning

After pretraining, the model is shown thousands of **prompt-response pairs** written by humans. It learns that when someone asks a question, the right behavior is to answer it clearly and helpfully — not just predict the statistically likely next word from the internet.

RLHF — Human Feedback as a Guide

Reinforcement Learning from Human Feedback (RLHF) takes this further. Human raters compare multiple model outputs and rank them. A "reward model" learns what humans prefer, and the main model is then trained to maximize those rewards — like a student refining their essays based on a professor's feedback loop.

The result: a model that's not just knowledgeable, but **aligned** — one that follows instructions, avoids harmful outputs, and communicates in a way humans find useful.

PART 4

Model Behavior & Limitations

Understanding what AI *can't* do reliably is just as important as knowing what it can. These aren't bugs to be fixed in the next version — many are fundamental properties of how these systems work.



Hallucinations: When AI Makes Things Up

AI models can generate text that sounds confident and polished — but is **completely wrong**. This is called a **hallucination**, and it's one of the most important limitations to understand.

Why does it happen?

The model is always predicting the next most plausible token — not retrieving verified facts. If it has never seen a clear answer in its training data, it will **interpolate** a confident-sounding one. It has no internal "I don't know" switch by default.

A real-world analogy

Imagine asking a very well-read friend about an obscure historical event. Rather than say "I'm not sure," they confidently fill in the gaps with plausible-sounding details. That's hallucination — fluent, coherent, and wrong.

What to do about it

Always verify high-stakes AI outputs. Use retrieval-augmented generation (RAG) to ground the model in actual documents. Treat AI outputs as a **first draft**, not a final source of truth.

Bias, Data Dependency & Probabilistic Outputs

Bias & Data Dependency

A model can only know what it was trained on. If the training data over-represents certain languages, cultures, or viewpoints, the model will reflect those skews. **Garbage in, garbage out** — but at civilizational scale. This is why data curation, diversity, and ongoing auditing matter enormously in responsible AI development.

Deterministic vs. Probabilistic

Unlike a calculator (same input → always same output), language models are **probabilistic**. A setting called **temperature** controls randomness. At temperature 0, the model always picks the highest-probability token — very predictable. At higher temperatures, it samples from a wider range — more creative, but potentially less accurate. This is why asking the same question twice can yield different answers.

Key Takeaways

🧩 Tokens & Prediction

AI generates language by predicting one token at a time, using everything in its context window as input

⚡ Transformers Win

Attention mechanisms let models understand relationships across long texts — something older architectures simply couldn't do

📚 Training = Reading + Feedback

Pretraining builds world knowledge; fine-tuning and RLHF shape helpful, aligned behavior

⚠️ Know the Limits

Hallucinations, bias, and probabilistic outputs are inherent — understanding them makes you a better AI practitioner

> Generative AI is extraordinarily powerful — and it's also fundamentally a pattern-completion engine. The best practitioners hold both of those truths at once.